Degradation Modeling Considering the Dependency of Rate and Volatility for Real-Time Prognostics With Error Correction

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***Abstract*— Degradation modeling and prognostics play an important role in improving system security and reducing maintenance costs. However, accurate degradation modeling for increasingly sophisticated equipment or components faces challenges of rationality and applicability caused by time-varying operational environments, complex and unknown intrinsic factors, and inappropriate sensor measurements. In this respect, this article proposes an improved Wiener process model (WPM) with adaptive drift by describing degradation rate-volatility-related effects (DRVREs), and develops the parameter estimation procedures based on the Kalman filter and expectation maximum (EM) algorithm. Then, the real-time remaining useful life (RUL) prediction in the form of distribution functions is derived. Next, considering the risk of prediction error accumulation caused by uncertainty and measurement bias, a framework composed of degradation modeling and real-time prognostics with error compensation is then established, where the Gaussian process regression (GPR) is utilized to correct the raw prognostics results. The presented approach realizes applicability while guaranteeing the model rationality. The effectiveness of the proposed framework in modeling degradation and improving the RUL prediction accuracy is validated through a simulation and applications to datasets from experimental battery cells and on-road electric vehicle (EV) batteries.**

***Index Terms*— Adaptive wiener process, degradation rate-volatility-related effects (DRVREs), error correction, Gaussian process regression (GPR), remaining useful life (RUL).**

I. Introduction

IN AN increasingly complex and interconnected world, timely and efficient maintenance of intricate equipment is not only economically advantageous but also crucial for operational safety [1], [2], [3]. Every component within complex systems, during its operating life, may undergo a gradual degradation process that progressively decreases its performance. Once the performance degradation exceeds a prefixed threshold, the component is considered end-of-life and requires timely maintenance. Otherwise, the equipment may malfunction and pose safety hazards, leading to economic losses as well as physical injuries. For instance, when the capacity of lithium batteries utilized in electric vehicles (EVs) deteriorates to 80% of the nominal value [4], batteries require replacement; otherwise, they are considerably vulnerable to fire and explosion [5]. Therefore, the current performance of components needs to be evaluated by real-time prognostics, i.e., predicting the remaining useful life (RUL) based on operational monitoring data, to guide the health management of the equipment. Through that, the system safety can be dramatically enhanced while losses are prevented and maintenance costs are reduced [6].

Conducting an appropriate RUL prediction requires modeling the most possible potential degradation process of a component's performance. Accurate degradation modeling and RUL prediction for sophisticated equipment or components involve significant challenges from the operational environment, intrinsic factors, and sensor measurements [7]. First, the environmental conditions and operational loads are generally time varying, which leads to the temporal uncertainty of degradation. The uncertainty is reflected in the fact that the degradation rate and volatility vary with operating conditions and are observed to be positively relevant [8], [9]. Second, intrinsic factors mainly include individual variation and the complex degradation mechanisms of components. The former means that the heterogeneity and randomness of degradation paths exist among different individuals [10], while the latter makes it difficult to develop applicable physical models. Third, measurement data may be not appropriate due to measurement bias, noise, and intercomponent interference, weakening the accuracy of RUL prediction [4].

To summarize the above three aspects, the degradation model is expected to capture the randomness, the uncertainty, the relevance between degradation rate and volatility, and nonlinearity. Furthermore, the degradation model ought to have low computational complexity and small data dependency while ensuring high accuracy, especially in the current application landscape [4]. To meet these demands, nowadays, a large quantity of techniques and methods have been developed and applied to the degradation modeling, which can be roughly classified into two categories, namely, statistical-model-based and data-driven methods [1], [4].

Statistical models represented by stochastic processes are naturally advantageous due to the ability to describe the stochasticity of the degradation and provide the probability

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density function (pdf) of RUL [1], [4]. In the category of stochastic processes, the Wiener process is the most widely used for degradation modeling because of its superiority in directly describing the degradation rate and volatility, and the uncertainty. Wiener process models (WPMs) have been extended to adapt to various operational conditions and degradation mechanisms [10], [11], [12], [13], [14]. However, most studies are unable to represent the relationship between degradation rate and volatility. To end this, several researchers established a new class of WPM by introducing a proportional relationship between the drift coefficient and diffusion coefficient [8], [9], [15], [16]. However, their studies only considered the linear assumption, which is inadequate for highly nonlinear degeneration behaviors.

WPMs offer strong mathematical interpretation and high extensibility [17]. However, simple models [8], [9], [10], [11] may prove imprecise, while complex models [18], [19] can be costly to compute, when faced with the dynamic degradation characteristics of intricate equipment. On the contrary, data-driven approaches have flexibility and adaptability to various characteristics [4]. For example, Gaussian process regression (GPR) [20], [21], [22] and relevance vector machine [23], [24] are utilized to fit intrinsic degradation modes. Novel deep learning methods, such as long short-term memory [25], [26], gated recurrent unit [27], and encoder–decoder [28], are modified and applied to extract the degradation from historical data. In practice, data-driven methods have been proven to be effective for modeling nonlinearity and complexity with high accuracy and low processing [29]. Meanwhile, the hybridization of data-driven methods has garnered increasing attention, recently, which combines more methods and provides more excellent results [30], [31], [32], [33], [34], [35].

However, the black-box nature of data-driven approaches results in low interpretability, incapability to reveal the degradation rate and volatility, and vulnerability to the lack of priori knowledge and historical data [36]. As to hybrid methods, high computational complexity is introduced, and professional knowledge is strongly needed, or even undesirable results are produced due to inappropriate combinations. Based on the above discussion, it can be found that there are almost no methods that can guarantee the model rationality, i.e., the ability to describe uncertainty, nonlinearity, and relevance and achieve applicability, i.e., low computational complexity, small data dependency, and high accuracy, simultaneously. Therefore, the effort of this article is to address these problems by extending the WPM and then proposing an innovative prognostics approach with error correction using GPR. The main works and contributions of this article are summarized as follows.

1) An improved adaptive WPM is proposed based on the formulation of the dependency between degradation rate and volatility. This proposed model can achieve rationality, reflecting the real degradation characteristics.

2) Parameter estimation procedures using the Kalman filter and expectation maximum (EM) algorithm for the improved adaptive WPM are developed to real-time update the model parameters. The procedures can characterize the time-varying uncertainty of degradation and overcome the strong dependence on historical data.

3) A real-time prognostics approach with error correction is constructed, where the explicit pdf of RUL could be derived and error-corrected by GRP, once the updated parameters are obtained. This approach appropriately combines the proposed WPM and GRP, which improves prediction accuracy while model rationality and low complexity remains.

The remainder of this article is arranged as follows. Section II presents the formulation of the improved adaptive Wiener process. Section III develops the real-time RUL prediction approach with error correction. Simulation and two case studies using data from experimental battery cells and on-road EV batteries are illustrated in Section IV to validate the presented method, and also some future works are discussed here. Finally, Section V gives the conclusion.

II. Adaptive Wiener Process Model Considering Dependency of Degradation Rate and Volatility

A. Basic Nonlinear WPM

Let the stochastic process $\{X(t), t \geq 0\}$ describe the cumulative degradation path of a specific component with a known initial value $X(0) = x_0$. In order to model the nonlinearity of degradation, a common nonlinear WPM is formulated as follows [10]:

$$X(t) = x_0 + \eta \int_0^t \varphi(\tau, \alpha) d\tau + \sigma B(t) \tag{1}$$

where η is the drift parameter reflecting the degradation rate, σ is the diffusion coefficient representing the degradation volatility, and $B(t)$ is the standard Brownian motion. Without loss of generality, we assume $x_0 = 0$, $\eta > 0$, and $\varphi(0, \alpha) = 0$. $\varphi(t, \alpha)$ is a monotonically increasing function of t with an unknown parameter vector α, which can be used to characterize the nonlinear performance of components. The model in (1) can cover linear and nonlinear paths through selecting $\varphi(t, \alpha)$ of different forms.

When the degradation $X(t)$ reaches a predetermined critical threshold ξ, the component is deemed failed. Hence, the lifetime T can be defined as the first passage time (FPT) of $X(t)$ crossing ξ, i.e.,

$$T = \inf\{t : X(t) \geq \xi | x_0 < \xi\}. \tag{2}$$

For the degradation model in (1), if $\varphi(t, \alpha)$ is a continuous function of time t, then the pdf of T can be given as [37] follows:

$$f_T(t) \cong \frac{1}{\sqrt{2\pi t}} \left(\frac{S(t)}{t} + \frac{\eta}{\sigma} \varphi(\tau, \alpha) \right) \exp\left(-\frac{S^2(t)}{2t} \right) \tag{3}$$

where $S(t) = (\xi - \eta \int_0^t \varphi(\tau, \alpha) d\tau)/\sigma$.

In most existing degradation models, σ is assumed to be constant. However, the degradation volatility of a component in practical engineering may be positively related to the degradation rate, which can be referred to as degradation rate-volatility-related effects (DRVREs). More specifically, if the component degrades faster, the volatility of the degradation over time could be also higher [8], [9]. Therefore, a constant diffusion assumption is restrictive to describe such phenomenon. In order to obtain a reasonable degradation model, we give the formulation of an improved adaptive WPM with DRVRE in the following.

B. Formulation of the Dependency

According to the physical implications of the drift parameter and diffusion coefficient, we describe the DRVRE by deducing the relationship between η and σ. For computational convenience and simplicity, we consider a power function form [10], [38] of $\varphi(t,\alpha)$ throughout this article, i.e., $\varphi(t,\alpha)=\alpha t^{\alpha-1}$.

From [15] and [39], the following proposition is introduced to formulate the DRVRE.

Proposition 1: For a degradation path which can be modeled by (1), if the failure mechanism of the component remains constant during the whole process of operation, then the following relationship can be deduced, i.e.,

$$\eta/\sigma^{2\alpha}=\rho \tag{4}$$

where ρ is a fixed parameter denoting the failure mechanism equivalence. On basis of the acceleration factor invariant principle [33], [34], [35], the complete proof of Proposition 1 can be seen in Appendix A.

From (4), we can see that the ratio of η and $\sigma^{2\alpha}$ remains constant under different stress levels. The precondition of (4), i.e., the invariance of the failure mechanism, is also an underlying assumption for prognostics. If the failure mechanism changes, it is impossible to predict the RUL precisely based on the historical data as the degradation trend may be not consistent.

C. Improved Adaptive WPM

Combined with the DRVRE and the basic model in (1), an improved WPM is proposed as follows:

$$X(t)=x_0+\eta t^{\alpha}+(\eta/\rho)^{1/2\alpha}B(t) \tag{5}$$

where σ is replaced by $(\eta/\rho)^{1/2\alpha}$. Let $\phi=[\eta,\rho,\alpha]$ be the model parameters. From (3), the conditional pdf of T for the proposed model is

$$f_T(t|\phi)\cong\frac{\xi+(\alpha-1)\eta t^{\alpha}}{\sqrt{2\pi t^3(\eta/\rho)^{1/\alpha}}}\exp\left(-\frac{\rho^{1/\alpha}(\xi-\eta t^{\alpha})^2}{2\eta^{1/\alpha}t}\right). \tag{6}$$

Considering that the real-time degradation measurement $X(t_n)=x_n$ is available, the degradation evolution based on the proposed model beyond time t_n can be given by

$$X(t+t_n)=x_n+\eta\Lambda(t)+(\eta/\rho)^{1/2\alpha}\Delta B(t) \tag{7}$$

where $\Lambda(t)=(t+t_n)^{\alpha}-t_n^{\alpha}$ and $\Delta B(t)=B(t+t_n)-B(t_n)$.

In order to derive the RUL distribution, we define a new stochastic process $Y(t)=X(t+t_n)-x_n$. Then, the RUL of the component at a time t_n can be equal to the FPT of $Y(t)$ with respect to the threshold $\xi-x_n$, i.e.,

$$R_n=\inf\{r_n:Y(r_n)\geq\xi-x_n|Y(0)=x_n\}. \tag{8}$$

Given the degradation observations up to time t_n, i.e., $\boldsymbol{x}_{1:n}=\{x_0,x_1,\ldots,x_n\}$, the pdf of the RUL conditional on ϕ, i.e., $f_{R_n}(r_n|\boldsymbol{x}_{1:n},\phi)$, can be given by the following proposition.

Proposition 2: If the current degradation observation sequence $\boldsymbol{x}_{1:n}$ and the parameter value ϕ are available, the pdf of the RUL at time t_n can be formulated as follows:

$$f_{R_n}(r_n|\boldsymbol{x}_{1:n},\phi)\cong\frac{\xi-x_n-\eta\left(\Lambda(r_n)-r_n\alpha(t_n+r_n)^{\alpha-1}\right)}{\sqrt{2\pi r_n^3(\eta/\rho)^{1/\alpha}}}\times\exp\left(-\frac{\rho^{1/\alpha}(\xi-x_n-\eta\Lambda(r_n))^2}{2\eta^{1/\alpha}r_n}\right). \tag{9}$$

The proof of Proposition 2 can be easily achieved by referring to the results in [10]. Before the RUL prediction, the model parameters and degradation state should be estimated. Therefore, we develop an adaptive model in the following.

To consider both the time-varying characteristics and stochastic of the degradation drift, η is assumed to be normally distributed to account for the uncertainty, and then, a state-space model based on the improved WPM is constructed as follows:

$$\begin{cases}\eta_i=\eta_{i-1}+\nu\Delta W_i\\ x_i=x_{i-1}+\eta_{i-1}\tau_i+(\eta_{i-1}/\rho)^{1/2\alpha}\Delta B_i\end{cases} \tag{10}$$

where ν is the diffusion coefficient of the drift parameter η, $\Delta W_i=W(t_i)-W(t_{i-1})$, $\Delta B_i=B(t_i)-B(t_{i-1})$, and $\tau_i=t_i^{\alpha}-t_{i-1}^{\alpha}$, and the two equations are, respectively, named the state equation and the observation equation.

In state equation, η is described by a random walk model with a Brownian motion $W(t)$ independent of $B(t)$, which can make it a time-varying random variable. Therefore, the drift can be updated adaptively once the current observed degradation data are available. Moreover, according to the DRERV relationship in Proposition 1, σ can be also updated by using $\sigma=(\eta/\rho)^{1/2\alpha}$. That is, both the drift parameter and the diffusion coefficient are adaptive in the proposed model, which is more suitable for real degradation processes.

III. Real-Time Prognostic With Error Correction

The real-time RUL can be predicted by using the state-space model proposed in Section II. A recursive filter algorithm is developed to update the drift parameter based on the whole historical degradation observations up to date. The unknown parameters of the proposed model are estimated by the EM algorithm. Then, the pdf of the real-time RUL can be predicted. In addition, considering that the risk of the RUL prediction error may increase as the degradation rate rises, an error predictive correction model is proposed to estimate the prediction error. Then, the final predictions can be obtained by the sum of the pdfs of RUL and error estimation.

A. Updating of the Adaptive Drift and Diffusion

Suppose that the initial drift parameter η_0 follows a Gaussian distribution with mean μ_0 and variance ς_0. According to the state-space model in (10), η_i also follows a Gaussian distribution. Let $\mu_{i|i}$ and $\varsigma_{i|i}$ be the mean and variance of $\eta_i|\boldsymbol{x}_{1:i}$, respectively. Then, $\mu_{i|i}$ and $\varsigma_{i|i}$ can be estimated by the following Kalman filter algorithm.

Step 1: Initialize μ_0 and ς_0.

Step 2: Estimate the state at time t_i

$$\begin{aligned}\mu_{i|i-1} &= \mu_{i-1|i-1}\\ \varsigma_{i|i-1} &= \varsigma_{i-1|i-1} + \nu^2(t_i - t_{i-1})\\ K_i &= \varsigma_{i|i-1}\tau_i\Big[\varsigma_{i|i-1}\tau_i^2 + \big(\mu_{i|i-1}/\rho\big)^{1/\alpha}(t_i - t_{i-1})\Big]^{-1}.\end{aligned}$$

Step 3: Update the mean and variance of η_i

$$\begin{aligned}\mu_{i|i} &= \mu_{i|i-1} + K_i\big(x_i - x_{i-1} - \mu_{i|i-1}\tau_i\big)\\ \varsigma_{i|i} &= \varsigma_{i|i-1} - K_i\varsigma_{i|i-1}\tau_i.\end{aligned}$$

Obviously, the whole historical degradation observations up to t_i can be captured via this filtering algorithm, and thus, $\mu_{i|i}$ and $\varsigma_{i|i}$ are updated iteratively. Based on the property of the Kalman filter, the pdf of $\eta_i|\boldsymbol{x}_{1:i}$ is still a Gaussian, i.e.,

$$p(\eta_i|\boldsymbol{x}_{1:i}) = \frac{1}{\sqrt{2\pi\varsigma_{i|i}}}\exp\left(-\frac{\big(\eta_i - \mu_{i|i}\big)^2}{2\varsigma_{i|i}}\right). \tag{11}$$

B. Adaptive Parameter Estimation

The other unknown model parameters should be also estimated for the proposed model to be used in real-time prognostic. Let $\boldsymbol{\theta} = [\mu_0, \varsigma_0, \rho, \alpha, \nu]$ be the parameter vector. Here, the estimates of η, σ, and $\boldsymbol{\theta}$ can be updated simultaneously when the new degradation observation is available.

Since the drift parameter η in the state-space model can be seen as a hidden variable, the complete data log-likelihood function for $\boldsymbol{x}_{1:n}$ can be given by

$$\begin{aligned}\ell(\boldsymbol{\theta}) &= \ln p\big(\boldsymbol{x}_{1:n}, \boldsymbol{\eta}_{1:n}|\boldsymbol{\theta}\big)\\ &= \ln p\big(\boldsymbol{x}_{1:n}\big|\boldsymbol{\eta}_{1:n}, \boldsymbol{\theta}\big) + \ln p\big(\boldsymbol{\eta}_{1:n}|\boldsymbol{\theta}\big) = \ln p(\eta_0|\boldsymbol{\theta})\\ &\quad + \sum_{i=1}^{n}\ln p(\eta_i|\eta_{i-1}, \boldsymbol{\theta}) + \sum_{i=1}^{n}\ln p(x_i|\eta_{i-1}, \boldsymbol{\theta})\end{aligned} \tag{12}$$

where $\boldsymbol{\eta}_{1:n} = \{\eta_0, \eta_1, \ldots, \eta_n\}$ is the updated drift sequence given the degradation observations $\boldsymbol{x}_{1:n}$. The pdfs of $\eta_i|\eta_{i-1}, \boldsymbol{\theta}$ and $x_i|\eta_{i-1}, \boldsymbol{\theta}$ can be easily derived from (10).

In the following, the steps of the EM algorithm are shown to estimate $\boldsymbol{\theta}$. The initial value of the parameter vector is set as $\boldsymbol{\theta}^{(0)}$, and the kth iterative result of the estimate conditional on $\boldsymbol{x}_{1:n}$ is denoted by $\boldsymbol{\theta}_n^{(k)}$.

E-Step: Given the current estimate $\boldsymbol{\theta}_n^{(k)}$ and the observations $\boldsymbol{x}_{1:n}$, the Q-function is then obtained by taking the conditional expectation of (12) as follows:

$$Q\big(\boldsymbol{\theta}\big|\boldsymbol{\theta}_n^{(k)}\big) = E_{\boldsymbol{\eta}_{1:n}|\boldsymbol{x}_{1:n},\boldsymbol{\theta}_n^{(k)}}\big[\ln p\big(\boldsymbol{x}_{1:n}, \boldsymbol{\eta}_{1:n}|\boldsymbol{\theta}\big)\big]. \tag{13}$$

The detailed computational procedures of (13) can be deduced by Rauch–Tung–Striebel smoothing algorithm [11].

M-Step: The estimate of the next iteration $\boldsymbol{\theta}_n^{(k+1)}$ can be obtained by maximizing the Q-function, i.e.,

$$\hat{\boldsymbol{\theta}}_n^{(k+1)} = \arg\max_{\boldsymbol{\theta}} Q\big(\boldsymbol{\theta}\big|\boldsymbol{\theta}_n^{(k)}\big). \tag{14}$$

By taking the first partial derivatives of (14) with respect to $\boldsymbol{\theta}$ and setting them to 0, we can obtain the estimate $\hat{\boldsymbol{\theta}}_n^{(k+1)}$.

Finally, the E-step and M-step are applied iteratively until the estimates converge, i.e., $|\boldsymbol{\theta}_n^{(k+1)} - \boldsymbol{\theta}_n^{(k)}| \le e$, where e is a preset convergence threshold.

C. Real-Time RUL Prediction

Once the model parameters are updated, the pdf of the RUL can be derived on the basis of (9) and (11). Since η is random and other model parameters are nonrandom, the pdf of the updated RUL conditional on $\boldsymbol{x}_{1:n}$ at time t_n can be obtained as follows:

$$f_{R_n}(r_n|\boldsymbol{x}_{1:n}) = \int f_{R_n|x_n,\phi}(r_n|\boldsymbol{x}_{1:n}, \phi)\,p(\eta_n|\boldsymbol{x}_{1:n})\mathrm{d}\eta_n. \tag{15}$$

Due to the existence of integral operation, it is hard to obtain the analytic expression of $f_{R_n}(r_n|\boldsymbol{x}_{1:n})$. Therefore, we propose an approximate RUL distribution. To avoid the lengthy derivation, Lemma 1 is first given as follows.

Lemma 1: If $z \sim N(a, b)$ and $g(z)$ are a continuous function which has the second-order derivative within the neighborhood of $z = a$, then the expectation and the variance of $g(z)$ can be, respectively, approximated by

$$E\big[g(z)\big] \approx g(a) + \frac{g''(a)}{2}b,\; D\big[g(z)\big] \approx \big[g'(a)\big]^2 b \tag{16}$$

where $g'(\cdot)$ and $g''(\cdot)$ denote the first-order derivative and the second-order derivative, respectively.

The proof of Lemma 1 can be obtained by using Taylor's formula. Then, based on Proposition 2 and Lemma 1, the following result is derived for the RUL.

Proposition 3: For the degradation process defined in (5) and given $\boldsymbol{x}_{1:n}$, the pdf of the RUL at time t_n can be approximated by

$$\begin{aligned}&f_{R_n}(r_n|\boldsymbol{x}_{1:n})\\ &\approx \exp\big[h_2\big(\mu_{n|n}\big)\big]\Bigg\{h_1\big(\mu_{n|n}\big)\\ &\quad + \frac{1}{2}\left[\frac{\partial^2 h_1(\eta_n)}{\partial\eta_n^2} + 2\frac{\partial h_1(\eta_n)}{\partial\eta_n}\frac{\partial h_2(\eta_n)}{\partial\eta_n}\right]\varsigma_{n|n}\\ &\quad + \frac{h_1\big(\mu_{n|n}\big)}{2}\left[\left(\frac{\partial h_2(\eta_n)}{\partial\eta_n}\right)^2 + \frac{\partial^2 h_2(\eta_n)}{\partial\eta_n^2}\right]\varsigma_{n|n}\Bigg\}_{\eta_n=\mu_{n|n}}\end{aligned} \tag{17}$$

where $h_1(\eta_n) = (\xi - x_n - \eta_n(\Lambda(r_n) - r_n\alpha(t_n + r_n)^{\alpha-1}))/(\sqrt{2\pi r_n^3(\eta_n/\rho)^{1/\alpha}})$ and

$$h_2(\eta_n) = \big[-\rho^{1/\alpha}(\xi - x_n - \eta_n\Lambda(r_n))^2\big]/\big(2\eta_n^{1/\alpha}r_n\big).$$

It can be seen that the whole historical degradation observations up to date are utilized in the prediction and the uncertainty of degradation are propagated through the iterative process of the drift.

D. Error Correction Model

Although the accuracy of the degradation model can be improved significantly, the error between the predicted RUL and the true RUL still exists due to the uncertainty and the measurement accuracy. Thus, we consider an error correction model (ECM) here to obtain more accurate predictions.

Note that the prediction errors at different time points may have different Gaussian distribution features because of the changing degradation rate and operation conditions. Hence, GPR is more suitable to model the time series of prediction errors. Based on the historical error data and GPR, the error of RUL prediction can be estimated. Then, the ECM is constructed for obtaining the final RUL prediction by correcting the predicted RUL with the error estimation.

Let $\boldsymbol{t}_{n-1} = [t_1, \ldots, t_{n-1}]^{\mathrm{T}}$ and $\boldsymbol{\varepsilon}_{n-1} = [\varepsilon_1, \ldots, \varepsilon_{n-1}]^{\mathrm{T}}$, where ε_{i} denotes the prediction error at time t_i, $i = 1, \ldots, n-1$. Assume that the error satisfies $\varepsilon_{\mathrm{i}} = q(t_i)$, where $q(\cdot)$ is a function of time. Let $\boldsymbol{q}(\boldsymbol{t}_{n-1}) = [q(t_1), \ldots, q(t_{n-1})]^{\mathrm{T}}$. According to the properties of GRP, $\boldsymbol{\varepsilon}_{n-1} = \boldsymbol{q}(\boldsymbol{t}_{n-1})$ follows a $n-1$ dimensional Gaussian distribution, i.e., $N(0, \boldsymbol{\Sigma})$. The covariance matrix is given by

$$\begin{aligned}\boldsymbol{\Sigma} &= \boldsymbol{\kappa}_{n-1,n-1} + \sigma_\varepsilon^2 \boldsymbol{I} \\ &= \begin{bmatrix} \kappa(t_1,t_1) & \kappa(t_1,t_2) & \cdots & \kappa(t_1,t_{n-1}) \\ \kappa(t_2,t_1) & \kappa(t_2,t_2) & \cdots & \kappa(t_2,t_{n-1}) \\ \vdots & \vdots & \ddots & \vdots \\ \kappa(t_{n-1},t_1) & \kappa(t_{n-1},t_2) & \cdots & \kappa(t_{n-1},t_{n-1}) \end{bmatrix} + \sigma_\varepsilon^2 \boldsymbol{I} \end{aligned} \quad (18)$$

where $\kappa(t_i, t_j)$ is the kernel function, σ_ε is the noise parameter, and $\boldsymbol{I}$ is the $n-1$ dimensional identity matrix. The squared exponential kernel is used here, i.e.,

$$\kappa\left(t_i, t_j\right) = \sigma_f^2 \exp\left(-\frac{\left(t_i - t_j\right)^2}{2\gamma^2}\right) \quad (19)$$

where σ_f and γ are the hyperparameters that can be determined by using the maximum likelihood estimation (MLE) method with historical data. Then, once the prediction errors up to data are available, and the prediction error of the next time point can be estimated by the following proposition.

Proposition 4: Given the prediction error sequence up to time t_{n-1}, $\boldsymbol{\varepsilon}_{n-1} = [\varepsilon_1, \ldots, \varepsilon_{n-1}]^{\mathrm{T}}$, and based on the properties of GPR, the distribution of the prediction error ε_n at time t_n can be estimated by

$$\hat{\varepsilon}_n | \boldsymbol{\varepsilon}_{n-1} \sim N\left(\boldsymbol{\kappa}_n \boldsymbol{\kappa}_{n-1,n-1}^{-1} \boldsymbol{\varepsilon}_{n-1}, \kappa_n - \boldsymbol{\kappa}_n \boldsymbol{\kappa}_{n-1,n-1}^{-1} \boldsymbol{\kappa}_n^{\mathrm{T}}\right) \quad (20)$$

where $\boldsymbol{\kappa}_n = [\kappa(t_n, t_1), \ldots, \kappa(t_n, t_{n-1})]^{\mathrm{T}}$ and $\kappa_n = \kappa(t_n, t_n)$. The proof of Proposition 4 can be seen in Appendix C.

After obtaining the estimation of the prediction error, the final RUL prediction with error correction at time t_n can be calculated as follows:

$$R_n^c = R_n + \varepsilon_n. \quad (21)$$

Note that the RUL prediction R_n and the error estimation ε_n are independent random variables. Hence, according to the convolution formula, the pdf of R_n^c can be deduced by

$$f_{R_n^c}\left(r_n^c\right) = \int_{-\infty}^{+\infty} f_{R_n}\left(r_n^c - \varepsilon_n | \boldsymbol{x}_{1:n}\right) p(\varepsilon_n | \boldsymbol{\varepsilon}_{n-1}) d\varepsilon_n. \quad (22)$$

The mean residual life (MRL) with error correction of the component at time t_n is formulated as follows:

$$\mathrm{MRL}_n^c = E\left[f_{R_n^c}\left(r_n^c\right)\right]. \quad (23)$$

In order to illustrate the framework of the proposed approach more intuitively, the corresponding flowchart is given in Fig. 1.

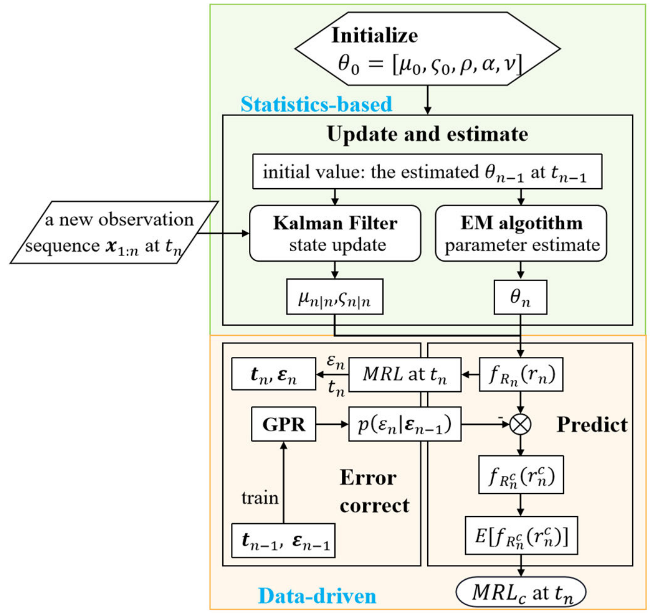


Fig. 1. Flowchart of the proposed approach.

IV. Illustrative Examples

In this section, a simulation study is first conducted to validate the rationality of the presented adaptive WPM considering DRVRE, and the effectiveness of the corresponding parameter estimation procedures. Then, the data from lithium-ion cells and EV batteries are analyzed to illustrate the superiority of the proposed RUL prediction approach combined with ECM.

A. Simulation Study

In this section, qualitative and quantitative results for both adaptive parameter estimation and one-step prediction of degradation path are presented, to, respectively, demonstrate the ability and benefits of our proposed method, defined as $\mathcal{M}_{\text{our}}$. The one-step prediction of degradation observation $x_{i+1|i}$ can be achieved by $x_{i+1|i} = x_i + \mu_{i|i}(t_{i+1}^\alpha - t_i^\alpha)$. In addition, two existing models are considered as comparison. One is proposed by Si [10], denoted by $\mathcal{M}_{\text{s}}$, a basic nonlinear Wiener process. The other model for the comparison is proposed by Wang et al. [15], denoted by $\mathcal{M}_{\text{w}}$ a linear Wiener process considering a fixed proportion relationship between η and σ. For illustration purposes, $\varphi(t, \alpha)$ of $\mathcal{M}_{\text{s}}$ is set to be the same as $\mathcal{M}_{\text{our}}$, while that of $\mathcal{M}_{\text{w}}$ is set to be t^0. Following the harmonization of the notation language, the details of these models are summarized in Table I:

$$\begin{cases} \eta_i = \eta_{i-1} + \nu \Delta W_i \\ x_i = x_{i-1} + \eta_{i-1}\left(t_i^\alpha - t_{i-1}^\alpha\right) + \sigma \Delta B_i \end{cases} \quad (24)$$

$$\begin{cases} \eta_i = \eta_{i-1} + \nu \Delta W_i \\ x_i = x_{i-1} + \eta_{i-1}(t_i - t_{i-1}) + \sqrt{\eta_{i-1}/\rho}\,\Delta B_i. \end{cases} \quad (25)$$

To begin with, the simulated degradation paths are sampled from the state-space model (10). We set the sampling number $T = 200$, sampling interval $\Delta t = 0.05$ s and model parameters $(\mu_0, \varsigma_0, \nu, \rho, \alpha) = (3, 0.01, 0.1, 0.2, 1.4)$. The simulation is repeated ten times, whose results are illustrated in Fig. 2.

TABLE I

DETAILS OF THREE MODELS

Details	$\mathcal{M}_{\mathrm{our}}$	$\mathcal{M}_{\mathrm{s}}$ [10]	$\mathcal{M}_{\mathrm{w}}$ [15]
Nonlinearity	++	++	-
DRVRE	++	-	+
Parameters	$\mu_0, \varsigma_0, \nu, \rho, \alpha$	$\mu_0, \varsigma_0, \nu, \sigma, \alpha$	$\mu_0, \varsigma_0, \nu, \rho$
State-space model	(10)	(24)	(25)

Notation: '++' means the model can describe the characteristic, while '+' means partial description and '-' means no description.

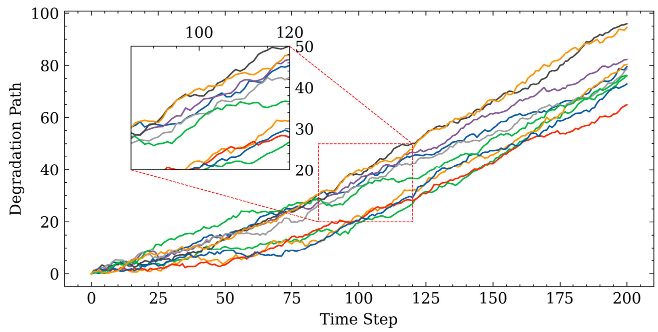


Fig. 2. Generated paths.

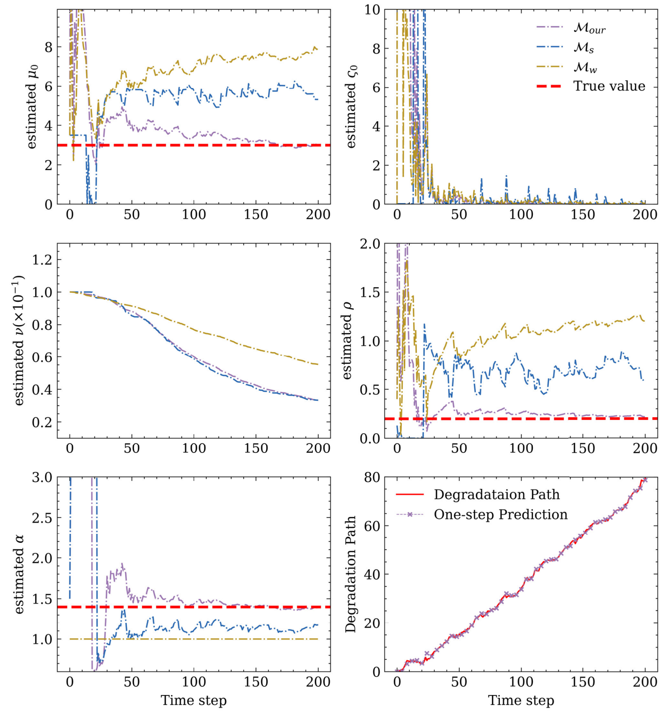


Fig. 3. Evolution of parameter estimation and one-step prediction.

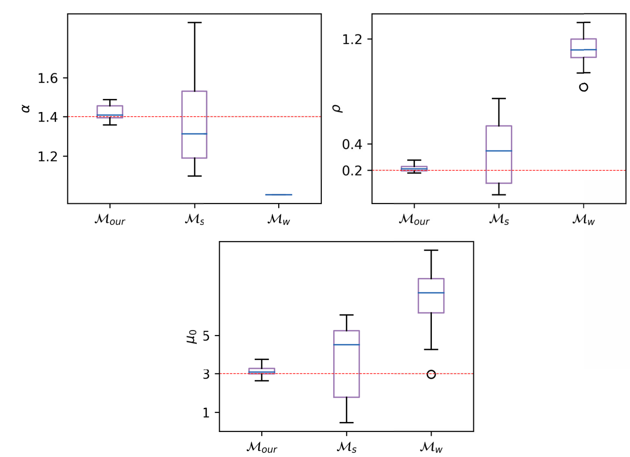


Fig. 4. Boxplots of final estimates.

We fit the paths using $\mathcal{M}_{\mathrm{our}}$, $\mathcal{M}_{\mathrm{w}}$, and $\mathcal{M}_{\mathrm{s}}$, separately, with the same random initialization. The operation procedures of $\mathcal{M}_{\mathrm{w}}$ and $\mathcal{M}_{\mathrm{s}}$ can be referred to [10] and [15].

For illustration purpose, the evolution of adaptive parameter estimation and one-step degradation prediction, by three models, for one of the simulated paths, are shown in Fig. 3. The final estimates of μ_0, ρ, and α for the ten paths are illustrated in box plots of Fig. 4. Note that, to contrast, the value of $\eta/\sigma^{2\alpha}$ calculated along with parameter updating is treated as the estimated ρ of $\mathcal{M}_{\mathrm{s}}$, and the estimated α of $\mathcal{M}_{\mathrm{w}}$ is considered as 1. From Figs. 3 and 4, it is obvious to see that only our model provides quite accurate and robust estimation of the values of μ_0, ρ, and α, indicating the excellent capacity of describing degradation characteristics.

In addition, the average of the mean absolute error (MAE) between the estimated and actual values of three parameters after the 15 time step in ten simulations is given in Table II, which also quantitatively validates the effectiveness of the parameter estimation procedures.

The reason why the estimated ν gradually declines is that the models tend to interpret the volatility as the degradation diffusion but not the fluctuations of degradation drift. As to the phenomenon that the estimated ς_0 is nearly 0 after convergence of other parameters, the estimated ς_0 reflects the smoothness of estimation of μ_0, decreasing with stabilization of the estimation. Last but not least, the MAE between one-step prediction $x_{i+1|i}$ and true degradation observation x_{i+1} for three models is also presented in Table II, providing another perspective to demonstrate the superiority of our model in degradation modeling.

TABLE II

AVERAGED MAE FOR TEN SIMULATIONS REGARDING DIFFERENT MODELS

MAE	$\mathcal{M}_{\mathrm{our}}$	$\mathcal{M}_{\mathrm{s}}$	$\mathcal{M}_{\mathrm{w}}$
α	0.128	0.404	-
ρ	0.044	0.426	0.771
μ_0	0.580	2.182	3.594
one-step prediction	0.453	0.503	0.487

B. Applications to Nonaccelerated Lithium-Ion Cells Data

In this case study, the proposed approach is applied to the lithium-ion cells for degradation modeling and prognostics. The dataset is provided by Lyu et al. [40]. It consists of eight nominally identical commercial lithium-ion batteries charge–discharge-cycled to a deeper failure, approximately 20% state of health (SOH) in an in-house environment without any

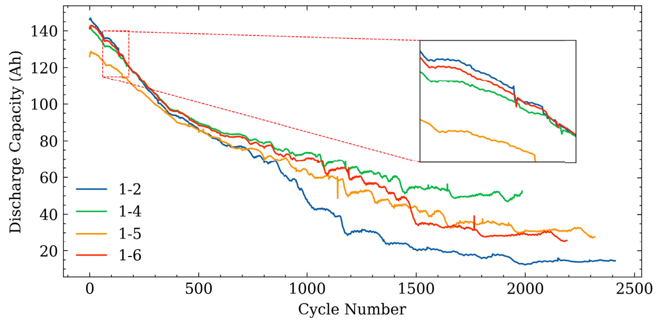


Fig. 5. Discharge capacity of cells 1-2, 1-4, 1-5, and 1-6.

control of temperature, where SOH is defined as follows [41]:

$$\text{SOH}=\frac{C_{\text{full}}}{C_{\text{nominal}}}\times 100\%$$

where C_{nominal} is the nominal capacity of the battery as a product and C_{full} is the full-charged discharge capacity after usage.

Meanwhile, the degradation experiments were conducted nonacceleratedly, since the cells were charged and discharged under nominal conditions. Fig. 5 shows half of the eight discharge capacity degradation paths, whose labels are cells 1-2, 1-4, 1-5, and 1-6, respectively.

As illustrated in Fig. 5, the experiments record a wide range of cycles. We select the portion of data where SOH is larger than 80%, since 80% SOH is considered as the threshold of end-of-life in practice [41]. The length of the extracted data is approximately between 150 and 250 cycles, and the RUL of a cell at the ith cycle is defined as the remaining cycle number, i.e., the length minus i. As to the data characteristics, nonlinearity, tiny fluctuations, and time-varying degradation rates can be seen in the degradation paths. Also, the degradation rate and volatility are altered synchronously driven by the change in environmental temperature [40]. These characteristics are suitable for the validation of the proposed method.

To evaluate our proposed prognostic framework with ECM, the models, $\mathcal{M}_{\text{s}}$ and $\mathcal{M}_{\text{w}}$, defined in the simulation part are still chosen as a comparison. To meet the general assumption of WPM, a transform of data is implemented by subtracting the initial value from the origin data and then taking the opposite, thereby making the initial value 0 and the degradation trend increasing. Then, the DRVRE in data is illustrated by performing a Pearson correlation test for η and $\sigma^{2\alpha}$. Obtained the corresponding updated parameters in estimation procedures of $\mathcal{M}_{\text{s}}$, the Pearson correlation coefficient of η and $\sigma^{2\alpha}$ can be calculated by

$$r_{\eta,\sigma^{2\alpha}}=\frac{\sum_i\left\{\left(\mu_{i|i}-\overline{\mu}\right)\left[\left(\sigma^{2\alpha}\right)_i-\overline{\sigma^{2\alpha}}\right]\right\}}{\sqrt{\sum_i\left(\mu_{i|i}-\overline{\mu}\right)^2}\sqrt{\sum_i\left[\left(\sigma^{2\alpha}\right)_i-\overline{\sigma^{2\alpha}}\right]^2}} \tag{26}$$

where $(\cdot)_i$ represents the updated value based on the data up to the ith cycle, while $\overline{(\cdot)}$ represents an average operation. The coefficient $r_{\eta,\sigma^{2\alpha}}$ and corresponding correlations for the four cells are given in Table III.

TABLE III
PEARSON CORRELATION COEFFICIENT OF η AND $\sigma^{2\alpha}$

Cell	1-2	1-4	1-5	1-6
$r_{\eta,\sigma^{2\alpha}}$	0.693328	0.849074	0.508839	0.915849
correlation	strong	strong	moderate	extremely strong

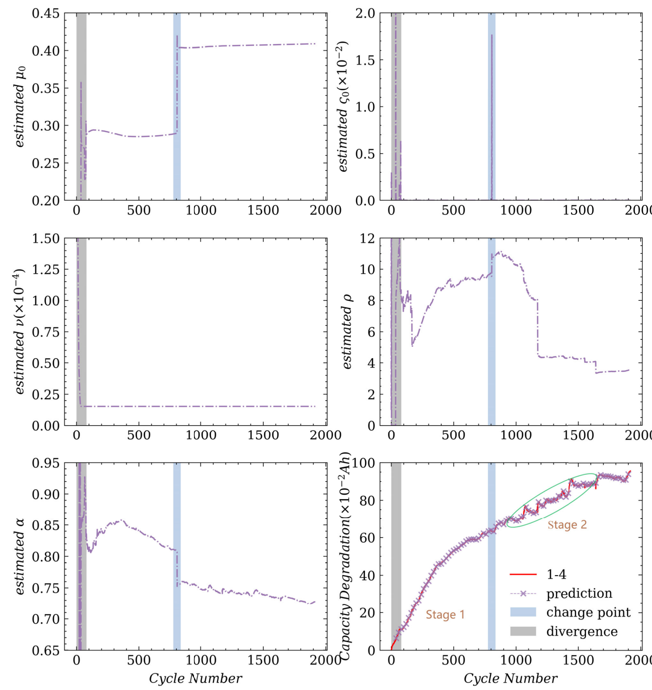
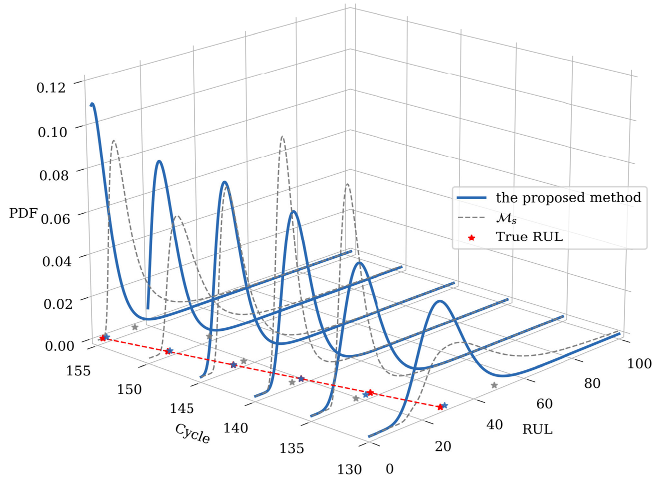


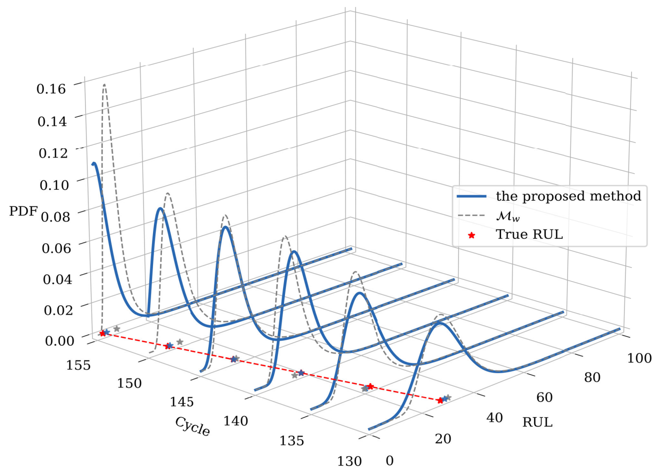
Fig. 6. Evolution of parameter estimation and one-step prediction.

From Table III, although the significance level of correlation varies from cell to cell due to individual variance, the results are sufficient to prove the proportional relationship between η and $\sigma^{2\alpha}$, i.e., DRVRE to be a rational hypothesis.

In the following, the data for cell 1-4 are used for illustration purposes, and especially, we extend the execution of the parameter estimation procedures to the whole data, while that of RUL prediction remains the data before 80% SOH. In addition, different from random initialization in the simulation study, the initial values for the illustrated cells are produced by the averaged results of estimation of the other four unillustrated ones. For cell 1-4, Fig. 6 displays the evolving process of the estimated parameters using our model. It can be found that both μ_0 and ν converge fast with the cycle number growing up. And, there is an obvious two-stage nature of the degradation, which can be reflected by the sudden changes in the estimated parameters. For instance, μ_0 and ς_0 mutate at the change point and then fall rapidly to new stable values. As to α, it increases first and then goes down at stage 1, while it gradually decreases at stage 2, which is consistent with the trend of nonlinearity in capacity degradation. In addition, the volatility of stage 2 is more obvious than that of stage 1, leading to a more stable estimation of ρ at stage 2. Taken together, the capability of our proposed model to describe nonlinearity and volatility, along with the adaptive parameter estimation procedures, can be demonstrated to be efficient.



(a)



(b)

Fig. 7. Predicted RUL based on different models. (a) Comparisons between $\mathcal{M}_{\text{our}}$ and $\mathcal{M}_{\text{s}}$. (b) Comparisons between $\mathcal{M}_{\text{our}}$ and $\mathcal{M}_w$.

The one-step predictions of capacity degradation are also illustrated in Fig. 6. It can be found that the presented model fits the degradation path very well and the MAE between the predictions and true degradation observations is 0.0520. In comparison, the MAEs of $\mathcal{M}_{\text{s}}$ and $\mathcal{M}_{\text{w}}$ are 0.0594 and 0.0556, respectively. This indicates that the proposed model has a higher accuracy in degradation prediction.

To compare the prognostic performance, the pdfs and MRLs of the RUL predicted by these methods are calculated based on the estimated parameters, separately. The true lifetime of cell 1-4 is considered as 157 cycles. The results of RUL prediction from the 130th to 155th cycle are shown in Fig. 7, where the MRL for a certain method is represented by the asterisk in its corresponding color of pdf. From Fig. 7, it can be found that the proposed approach performs well in RUL prediction, providing the closest MRLs to the true RULs. The absolute expectation error (AEE) between the predicted and true RUL at each cycle is given in Fig. 8. to quantitatively illustrate the superiority.

Also, the raw results of our model without applying ECM, marked by $\mathcal{M}_{\text{our}}^{-}$, are provided to show the efficacy of error

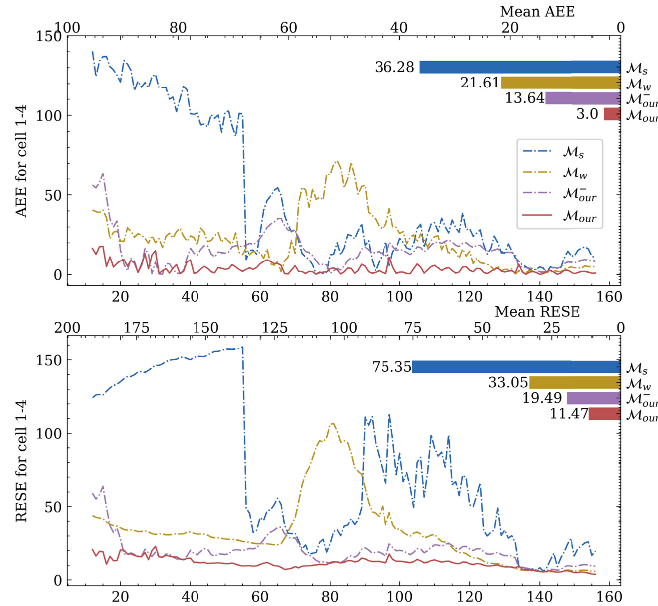


Fig. 8. AEEs and RESEs of prediction at every cycle.

TABLE IV
RESULTS OF PREDICTION

Cell	1-6		1-5		1-2	
	MAEE	MRESE	MAEE	MRESE	MAEE	MRESE
$\mathcal{M}_{\text{s}}$	39.55	23.61	50.24	27.89	53.49	41.37
$\mathcal{M}_{\text{w}}$	27.99	16.71	37.84	29.20	48.92	44.70
$\mathcal{M}_{\text{our}}^{-}$	22.48	15.12	20.78	18.19	39.85	23.21
$\mathcal{M}_{\text{our}}$	8.21	12.95	9.42	17.45	16.67	18.77

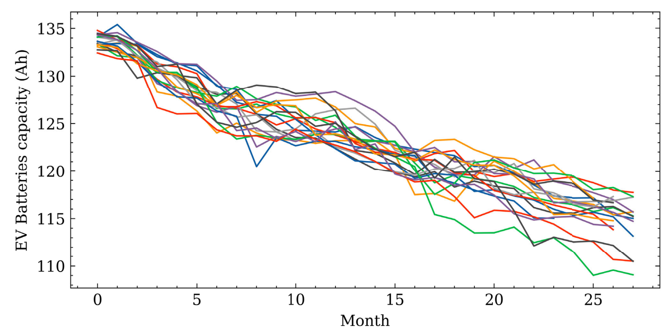


Fig. 9. Monthly averaged capacity of EV batteries within 29 months.

correction. In addition, a new metric, root expectation square error (RESE) at each cycle is introduced to describe the dispersion of the predicted RUL described by a pdf from the actual RUL. The formulas of AEE and RESE can be seen in Appendix B. From Fig. 8, it is intuitive to see that ECM contributes positively to the prediction accuracy at most points, which enlarges the advantages of our improved model considering DRVRE in prognostics over the models that do not consider DRVRE or nonlinearity. And, the advantages are twofold. First, it provides a more precise point estimate, and second, the estimated distribution has less dispersion from the true value.

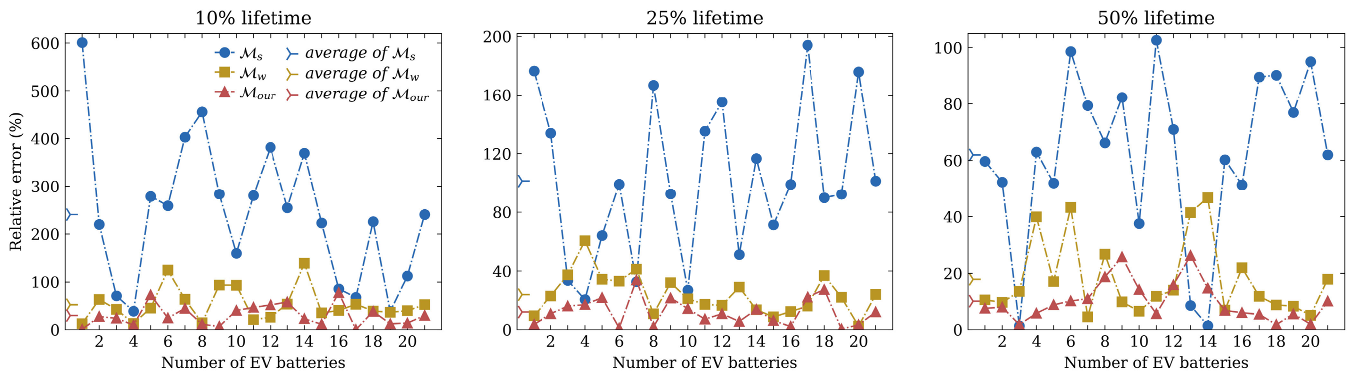


Fig. 10. RAEEs of the predicted RUL and the corresponding average for each EV battery.

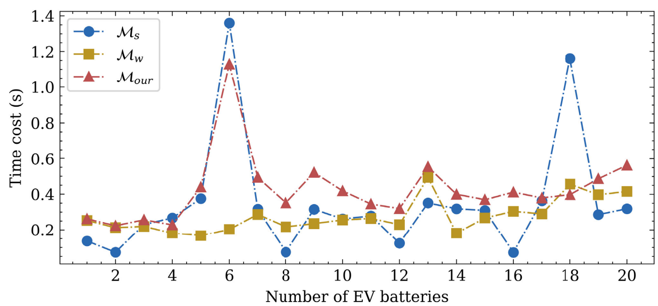


Fig. 11. Averaged time cost to process a new degradation observation, using Python 3.9 on a laptop with Intel i9-12900H CPU.

Note that, the two errors of $\mathcal{M}_s$ are particularly evident at early cycles, since it does not converge at that stage.

These methods are also applied to cells 1-2, 1-5, and 1-6, where the mean AEE (MAEE) and mean RESE (MRESE) of RUL prediction are provided in Table IV. Similar to cell 1-4, our proposed method provides the prediction with the lowest errors on average. These results demonstrate the excellent diagnostic performance of our method when implemented on battery data from nonaccelerated charge–discharge cycle tests.

C. Applications to EV Batteries Data

To further illustrate the applicability of the proposed method, its implementation on the EV battery data from real-life scenarios is also presented. The dataset is provided by Deng et al. [42]. It records the charging profiles of 20 EVs over a period of 29 months, including charging time, charging current, and state of charge during every charging process. The actual battery capacity for an EV once charged can be calculated by the Ampere integral formula [42]. Instead of the number of charges, the actual months are selected as the time unit for prognostics, where the averaged calculated capacity during a month represents the true capacity of this month.

Fig. 9 illustrates the capacity degradation over months. The same procedures as Section IV-B are applied to every EV battery. The prognostic performance of all the batteries is provided in Fig. 10, which contains the relative AEEs (RAEE) of the predicted RUL at 10%, 25%, and 50% of the battery lifetime left.

As expected, our proposed approach performs the best in nearly all EV batteries at different levels of remaining lifetime and gives the minimal averaged relative errors. Instead, $\mathcal{M}_s$ fails to provide robust results with acceptable error. These results in Fig. 10 indicate that our method is capable of performing the prognostics task with high accuracy in practical scenarios and even small sample situations. In addition, the comparisons between the computational time costs of different methods are provided in Fig. 11, where the average time of processing a new degradation observation is shown. It can be seen that our model is slightly more time consuming due to more details. But overall, the taken time is on the same order of magnitude as that of other models. In summary, our proposed method apparently improves accuracy with a little additional time consume, representing the applicability of our prognostics framework with ECM.

D. Discussion

Our proposed approach enriches the WPM family by considering DRVRE in nonlinear degradation and also makes a useful attempt to correct prediction error and improve accuracy by combining a machine learning algorithm. Meanwhile, there is space for further improvement in the following aspects.

1) In our study, only the systematic errors caused by diffusion but not the observation error are considered in the observation equation. Although the accuracy loss caused by the latter can be compensated by ECM, what effect it would have on the judgment of the correlation between the drift and diffusion is worth studying.

2) It is worth trying to apply our model to massive battery data from an identical batch under identical working conditions. Since these batteries are more likely to have similar lifetimes and degradation characteristics, the estimated results from batteries for training can be used as the initial value in the state estimation of those for testing, like we do, and also, ECM will be more scalable, which can provide a more comprehensive and generalized conclusion.

V. Conclusion

In this study, a complete framework of degradation modeling and real-time prognostics is proposed, including an RUL prediction method based on an improved WPM, adaptive state updating and parameter estimation procedures, and an ECM. This framework is designed to provide higher modeling accuracy for predicting the RUL of components with considering nonlinearity DRVRE and time-dependency uncertainty characteristics.

Our works can be summarized as follows.

1) In order to model the degradation rationally, the nonlinear WPM is selected as the basis to describe the nonlinearity and uncertainty, the formulation of the dependency between η and σ is derived in detail to describe the DRVRE, and then, the improved adaptive WPM considering DRVRE is developed.

2) The explicit pdf of real-time RUL is expressed to predict the lifespan after gaining the updated parameters through the developed estimation procedures, and the ECM based on GRP is considered to compensate for accumulated potential prediction errors.

3) Last but not least, the ability to capture nonlinearity and correlation, and predicting the degradation is emphasized in the simulation part. Furthermore, the advantages of our proposed approach compared with existing models with respect to prediction accuracy are fully demonstrated by using datasets of experimental cells and actual EV batteries in the two case studies. In conclusion, the superiority of the proposed method in terms of the degradation modeling and the RUL prediction is verified through the comparison between different models in the simulation and case study.

Appendix

A. Proof of Proposition 1

Suppose that the component is performed under two different operation stresses, i.e., the kth and lth accelerated stress levels. The cumulative distribution functions (cdfs) of the lifetime under these two stress levels are denoted by $F_k(t_k)$ and $F_l(t_l)$, respectively. If $F_k(t_k) = F_l(t_l)$, the acceleration factor $\mathrm{AF}_{k,l}$ is defined as follows:

$$\mathrm{AF}_{k,l} = t_l/t_k. \tag{27}$$

Based on the acceleration factor constant principle, $\mathrm{AF}_{l,q}$ should be a constant. That is, the acceleration factor is determined only by the stress levels if the failure mechanism remains unchanged. Note that, the invariance of the failure mechanism of components under a stable operation environment is always holden. Hence, for any time t_k, $t_l > 0$, the following equation should be satisfied, i.e.:

$$F_k(t_k) = F_l\left(\mathrm{AF}_{k,l} t_k\right). \tag{28}$$

Taking the partial derivative of (28) with respect to t_k yields

$$f_k(t_k) = \mathrm{AF}_{k,l} \cdot f_l\left(\mathrm{AF}_{k,l} t_k\right) \tag{29}$$

where $f_k(t_k)$ and $f_l(t_l)$ are the PDFs of the lifetime under the kth accelerated stress level and the lth accelerated stress level, respectively.

Substitute (3) into (29), and we can have

$$\begin{aligned}\mathrm{AF}_{k,l} &= \frac{f_k(t_k)}{f_l\left(\mathrm{AF}_{k,l} t_k\right)}\\ &= \mathrm{AF}_{k,l}^{\frac{3}{2}} \frac{\sigma_l}{\sigma_k} D_1 \exp\left(\frac{\xi^2}{t_k} D_2 + t_k^{\alpha-1} \xi D_3 + t_k^{2\alpha-1} D_4\right)\end{aligned} \tag{30}$$

where $D_1 = (\xi + (\alpha-1)\eta_k t_k^{\alpha})/(\xi + (\alpha-1)\eta_l t_k^{\alpha} \mathrm{AF}_{k,l}^{\alpha})$, $D_2 = (1)/(2\sigma_l^2 \mathrm{AF}_{k,l}) - (1)/(2\sigma_k^2)$, $D_3 = (\eta_k)/(\sigma_k^2) - (\eta_l)/(\sigma_l^2)\mathrm{AF}_{k,l}^{\alpha-1}$, and $D_4 = (\eta_l^2)/(2\sigma_l^2)\mathrm{AF}_{k,l}^{2\alpha-1} - (\eta_k^2)/(2\sigma_k^2)$.

To ensure $\mathrm{AF}_{k,l}$ is a constant with any time t_k, the terms in (30) must satisfy the equations, $D_1 = 1$ and $D_2 = D_3 = D_4 = 0$. Thus, the following relationships can be derived

$$\mathrm{AF}_{k,l} = \frac{\sigma_k^2}{\sigma_l^2} = \frac{\eta_k^{1/\alpha}}{\eta_l^{1/\alpha}}. \tag{31}$$

Then, we can obtain the result in (4).

B. Proof of Proposition 3

Note that (15) can be rewritten as follows:

$$f_{R_n}(r_n|\boldsymbol{x}_{1:n}) = E_{\eta_n|\boldsymbol{x}_{1:n}}\left[f_{R_n}(r_n|\boldsymbol{x}_{1:n}, \eta_n)\right] \tag{32}$$

where $f_{R_n}(r_n|\boldsymbol{x}_{1:n}, \eta_n) = f_{R_n}(r_n|\boldsymbol{x}_{1:n}, \phi)$ and other nonrandom parameters are omitted in the conditional part.

From (11), $\eta_n|\boldsymbol{x}_{1:n}$ is the Gaussian distributed with mean $\mu_{n|n}$ and variance $\varsigma_{n|n}$. As a result of Lemma 1, $f_{R_n}(r_n|\boldsymbol{x}_{1:n})$ can be approximated by

$$\begin{aligned}f_{R_n}(r_n|\boldsymbol{x}_{1:n}) &\approx f_{R_n}\left(r_n|\boldsymbol{x}_{1:n}, \eta_n = \mu_{n|n}\right)\\ &\quad + \frac{1}{2}\left[\frac{\partial f^2_{R_n|x_n,\phi}(r_n|\boldsymbol{x}_{1:n}, \eta_n)}{\partial \eta_n^2}\Big|_{\eta_n=\mu_{n|n}}\right] \cdot \varsigma_{n|n}\end{aligned} \tag{33}$$

which is the result in (17).

C. Proof of Proposition 4

Based on the GPR, $\boldsymbol{\varepsilon}_{n-1}$ follows the Gaussian distribution $N(0, \boldsymbol{\Sigma})$. Then, we can find that $[\boldsymbol{\varepsilon}_{n-1}^{\mathrm{T}}, \varepsilon_n]^{\mathrm{T}}$ also follows a Gaussian distribution, i.e.,

$$\begin{bmatrix}\boldsymbol{\varepsilon}_{n-1}\\ \varepsilon_n\end{bmatrix} \sim N\left(0, \begin{bmatrix}\boldsymbol{\kappa}_{n-1,n-1} & \boldsymbol{\kappa}_n\\ \boldsymbol{\kappa}_n^{\mathrm{T}} & \kappa_n\end{bmatrix}\right). \tag{34}$$

Given the prediction error $\boldsymbol{\varepsilon}_{n-1} = [\varepsilon_1, \ldots, \varepsilon_{n-1}]^{\mathrm{T}}$, the conditional pdf of ε_n can be derived as follows:

$$\begin{aligned}p(\varepsilon_n|\boldsymbol{\varepsilon}_{n-1}) &= p(\varepsilon_n, \boldsymbol{\varepsilon}_{n-1})/p(\boldsymbol{\varepsilon}_{n-1})\\ &\sim N\left(\boldsymbol{\kappa}_n \boldsymbol{\kappa}_{n-1,n-1}^{-1} \boldsymbol{\varepsilon}_{n-1}, \kappa_n - \boldsymbol{\kappa}_n \boldsymbol{\kappa}_{n-1,n-1}^{-1} \boldsymbol{\kappa}_n^{\mathrm{T}}\right).\end{aligned} \tag{35}$$

This completes the proof.

D. Metrics

$$\begin{aligned}\mathrm{AEE}_i &= |E\left(r_i - \mathrm{RUL}_i^{\mathrm{true}}\right)|\\ &= |\int \left(r_i - \mathrm{RUL}_i^{\mathrm{true}}\right) f_{R_i}(r_i|\boldsymbol{x}_{1:i}) \mathrm{d}r_i|\\ &= |\mathrm{MRL}_i - \mathrm{RUL}_i^{\mathrm{true}}|\\ \mathrm{RESE}_i &= E\left[\left(r_i - \mathrm{RUL}_i^{\mathrm{true}}\right)^2\right]\\ &= \int \left(r_i - \mathrm{RUL}_i^{\mathrm{true}}\right)^2 f_{R_i}(r_i|\boldsymbol{x}_{1:i}) \mathrm{d}r_i\\ \mathrm{RAEE}_i &= \frac{\mathrm{AEE}_i}{\mathrm{RUL}_i^{\mathrm{true}}} = \frac{|\mathrm{MRL}_i - \mathrm{RUL}_i^{\mathrm{true}}|}{\mathrm{RUL}_i^{\mathrm{true}}}.\end{aligned}$$

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